

Diffusion-Driven Graph Reconstruction and Cross-View Contrastive Learning for Bundle Recommendation

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Abstract

Bundle recommendation integrates bundle-level feedback and item-level information to model users’ preferences for sets of items. Despite advances in multi-view modeling, two challenges remain: (1) noisy user-bundle (U-B) interactions can introduce misleading signals into graph propagation, distorting the representations used to learn bundle preferences; (2) sparse U-B feedback provides limited evidence for personalized preference learning, making it difficult to characterize users with few historical bundle interactions. To address these challenges, we propose a diffusion-driven graph reconstruction and cross-view contrastive learning framework for bundle recommendation. First, we develop a U-B graph denoising pipeline that integrates pretraining, user-conditioned latent diffusion, and graph reconstruction. Starting from pre-trained user and bundle embeddings, we refine each user’s aggregated historical bundle representation through conditional diffusion denoising and use it to re-score observed interactions and reconstruct the U-B graph. Within this pipeline, we tailor the denoising objective to bundle recommendation through a bundle-set consistency loss that encourages the recovered preferences to retain the semantics of the user’s historical bundle set. Second, we introduce pre-fusion cross-view contrastive learning anchored in the reconstructed U-B view. User-side and bundle-side objectives connect this view to the user-item and bundle-item views before representation fusion, enabling auxiliary information to strengthen preference learning under sparse U-B feedback. Experiments on three public benchmarks show that the proposed framework outperforms the compared baselines. Ablation and interaction-selection analyses support the proposed designs, while the user-group analysis shows the largest relative gains from the cross-view objective for users with the fewest bundle interactions.

1 Introduction

Users often seek a coherent collection of items rather than a single product. A coordinated outfit or a themed playlist, for example, can satisfy a need that would otherwise require users to assemble suitable items themselves. Bundle recommendation addresses this demand by matching users with sets of related items (Ma et al. 2022; Sun et al. 2024b). Unlike

individual-item recommendation, it must capture preferences for a bundle as a whole while accounting for its constituent items. We focus on ranking predefined bundles using historical interactions and bundle composition information.

This setting involves three relations with distinct roles. User-bundle (U-B) interactions record users’ choices of complete bundles and provide direct evidence of bundle preferences. User-item (U-I) interactions reflect preferences for individual items, whereas bundle-item (B-I) affiliations describe how items constitute bundles. Representations learned from these relations offer complementary views of users and bundles: the U-B view captures bundle-level collaborative behavior, while the U-I and B-I views contribute item-level behavioral and compositional information (Chen et al. 2019; Chang et al. 2020; Ma et al. 2024). Learning across these views allows a recommender to exploit information beyond users’ observed bundle choices.

Early approaches use attention to aggregate constituent items and jointly learn from bundle and item interactions (Chen et al. 2019). Graph neural networks (GNNs) extend this modeling to higher-order connectivity among users, bundles, and items (Chang et al. 2020). More recently, contrastive learning (CL) has provided a mechanism for connecting different views. CrossCBR encourages agreement between bundle-level and item-level representations through cross-view objectives (Ma et al. 2022). MultiCBR separately models all three relations, fuses their representations, and applies contrastive learning after fusion (Ma et al. 2024). These approaches demonstrate the value of multi-view information. Nevertheless, learning bundle preferences still faces two challenges arising from noise in observed U-B interactions and limited feedback from individual users.

Challenge 1: Noise in user-bundle interactions. Implicit feedback can include observations that do not faithfully reflect user preferences (Wang et al. 2021). When such interactions enter the U-B graph, their influence may spread through neighborhood aggregation and affect the representations used for bundle ranking (Wu et al. 2021). The difficulty lies in reassessing individual connections in the context of a user’s historical bundle set. Each edge records an interaction with one bundle, whereas the representations of multiple interacted bundles jointly provide evidence about the user’s bundle preferences. Simply reducing the number of edges does not determine which connections should be retained.

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The challenge is to exploit this personalized historical context to suppress noisy connections while preserving relevant bundle-preference information.

Challenge 2: Sparse user-bundle feedback. Users with few bundle interactions provide limited evidence for learning personalized bundle preferences. U-I behavior and B-I composition offer complementary information, but their relevance to bundle choice must be learned: liking individual items is not equivalent to preferring a complete bundle, and bundle composition alone does not establish a user’s interest. CrossCBR aligns bundle-view and item-view representations through cross-view contrastive objectives, whereas DGMAE transfers first- and higher-order U-B knowledge to item-view representation learning through distillation (Ma et al. 2022; Ren et al. 2023). Their reported effectiveness demonstrates the value of linking bundle-level evidence with item-level behavioral and compositional information. The challenge is therefore to establish explicit associations between bundle-level and auxiliary-view representations, allowing item-level information to support preference learning when direct U-B feedback is limited.

To address these challenges, we propose **ReCoBR** (Reconstruction and Contrastive Learning for Bundle Recommendation), a unified framework that combines diffusion-driven graph reconstruction with cross-view contrastive learning. **First, we develop a U-B graph denoising framework that integrates pretraining, user-conditioned latent diffusion, and graph reconstruction.** We pretrain user and bundle embeddings on observed U-B interactions, then aggregate each user’s historical bundle embeddings into a latent preference vector. Following the diffusion denoising formulation (Ho, Jain, and Abbeel 2020; Wang et al. 2023), we perturb this vector with progressively increasing noise and learn to recover it using the pretrained user embedding as a personalized condition. Within this framework, we adapt the denoising objective to bundle recommendation by introducing a bundle-set consistency loss (SetLoss). This constraint encourages the recovered preference vector to remain semantically consistent with the user’s historical bundle representations. By limiting drift from this bundle-level preference context during recovery, SetLoss helps preserve preference-relevant information for subsequent interaction re-scoring and graph reconstruction. We then use the refined preference vector to re-score observed interactions and select connections for reconstructing the U-B bipartite graph. The complete pipeline thus turns personalized preference denoising into a refinement of the graph used for recommendation.

Second, we introduce pre-fusion cross-view contrastive learning anchored in the reconstructed U-B view to address sparse bundle feedback. The recommendation backbone learns user and bundle representations from the reconstructed U-B graph and the U-I and B-I relations, then fuses the view-specific representations for prediction. Before fusion, we construct U-B–U-I and U-B–B-I contrastive pairs on both the user and bundle sides, using the reconstructed U-B view as their common anchor. These objectives encourage agreement between representations of the same entity across paired views while distinguishing different entities. By explicitly connecting auxiliary information to bundle-level rep-

resentations, they provide additional learning signals beyond each user’s limited U-B history. The two modules therefore have distinct but connected roles: the first refines the U-B connections used for propagation, and the second uses representations learned from that graph to support preference learning across views.

Our main contributions are as follows:

- We propose a diffusion-driven U-B graph denoising framework that integrates pretraining, user-conditioned latent diffusion, and interaction selection to reconstruct the graph used for recommendation. Within this framework, we adapt the denoising objective to bundle recommendation through a bundle-set consistency loss that encourages semantic consistency with historical bundle preferences.
- We introduce pre-fusion cross-view contrastive learning anchored in the reconstructed U-B view on both the user and bundle sides. By connecting U-B representations with U-I and B-I representations, this design exploits auxiliary information to alleviate the learning difficulties caused by sparse bundle feedback.
- Experiments on three public benchmarks demonstrate ReCoBR’s effectiveness through baseline comparisons, ablations, and interaction-selection analyses. A user-group analysis further shows that the proposed cross-view objective yields the largest relative gains for the lowest-interaction group.

2 Related Work

2.1 Bundle Recommendation

Bundle recommendation includes both ranking existing bundles and generating new item combinations (Sun et al. 2024a). For predefined-bundle ranking, a central issue is how to combine bundle-level interaction evidence with information about constituent items. DAM addresses this issue through attentive item aggregation and joint learning of U-B and U-I interactions (Chen et al. 2019). BGCN uses graph propagation to capture bundle-level and item-level collaborative information (Chang et al. 2020). In the broader collaborative filtering setting, LightGCN demonstrates the effectiveness of simplified neighborhood aggregation (He et al. 2020). Subsequent bundle models enrich the preferences and composition patterns captured by graph representations. MIDGN disentangles user intents across global and local views (Zhao et al. 2022), and BundleGT uses a hierarchical graph transformer to model bundling strategies (Wei et al. 2023).

Beyond representation construction, another line of research focuses on cooperation across views. CrossCBR contrasts bundle-view and item-view representations to encourage mutual enhancement (Ma et al. 2022). MultiCBR explicitly models B-I relations alongside U-B and U-I relations, and combines representation fusion with post-fusion contrastive learning (Ma et al. 2024). DGMAE instead distills first- and higher-order U-B knowledge into an item-view graph masked autoencoder (Ren et al. 2023). These studies establish several ways to exploit auxiliary information for bundle preference learning. ReCoBR builds on this multi-view setting by refining the U-B graph through conditional latent diffusion and

using the reconstructed view to anchor cross-view learning before fusion.

2.2 Denoising and Diffusion for Recommendation

Interaction denoising seeks to reduce the influence of observed feedback that does not accurately reflect user preferences. Adaptive Denoising Training identifies potentially noisy feedback through training losses and reduces its influence using loss truncation or reweighting (Wang et al. 2021). Diffusion models provide another way to learn from corrupted inputs by reversing controlled perturbations (Ho, Jain, and Abbeel 2020). DiffRec applies this formulation to interaction-vector reconstruction and introduces a latent-space variant for efficiency (Wang et al. 2023). DiffuRec operates on target-item embeddings and uses interaction sequences to condition their recovery for next-item prediction (Li, Sun, and Li 2023). These methods illustrate that the denoising object and conditioning information must be chosen to suit the recommendation task.

Diffusion has also been applied to bundle recommendation. DCBR combines conditional U-B interaction denoising with disentangled contrastive learning (Li 2025). RDiffBR uses residual diffusion on item-level bundle embeddings to accommodate changes in bundle composition (Zhang et al. 2026). ReCoBR instead focuses on the connection between historical bundle-preference modeling and U-B graph reconstruction. It uses pretrained user representations to condition the denoising of aggregated historical bundle embeddings, and translates the refined preferences into observed-edge selection. Within this pipeline, the bundle-set consistency loss adapts the denoising objective to historical bundle semantics, supporting the reconstruction of the graph used for downstream learning.

2.3 Contrastive Learning for Recommendation

Contrastive learning derives learning signals by encouraging agreement between related representations while distinguishing unrelated ones. SimCLR demonstrates the importance of view construction and contrastive objectives for representation learning (Chen et al. 2020). For graph-based recommendation, SGL creates perturbed graph views through operations such as edge and node dropout, then contrasts representations of the same node across views (Wu et al. 2021). Knowledge-aware methods extend this principle to structured auxiliary information. KGCL uses consistency under knowledge-graph perturbations to guide interaction augmentation (Yang et al. 2022), while MCCLK connects collaborative, semantic, and structural representations through multi-level cross-view objectives (Zou et al. 2022).

The bundle-specific methods discussed above apply contrastive learning to views defined by different user, bundle, and item relations (Ma et al. 2022, 2024). ReCoBR follows this approach with a targeted choice of contrastive pairs: U-B-U-I and U-B-B-I for both users and bundles, formed before representation fusion. The reconstructed U-B view provides a common reference for jointly optimizing the paired representations. This links item-level behavioral and compositional information to bundle-level preference learning,

helping the model exploit auxiliary evidence under sparse U-B feedback.

3 Methodology

3.1 Task Definition and Framework Overview

Let $\mathcal{U}$, $\mathcal{B}$, and $\mathcal{I}$ denote the sets of users, predefined bundles, and items, respectively. The available training data comprise three binary matrices: user-bundle interactions $\mathbf{X} \in \{0, 1\}^{|\mathcal{U}| \times |\mathcal{B}|}$, user-item interactions $\mathbf{Y} \in \{0, 1\}^{|\mathcal{U}| \times |\mathcal{I}|}$, and bundle-item affiliations $\mathbf{Z} \in \{0, 1\}^{|\mathcal{B}| \times |\mathcal{I}|}$. An entry equal to one indicates an observed interaction or an item belonging to a bundle. Given these relations, our task is to estimate a preference score $\hat{y}_{ub}$ for each unobserved user-bundle pair and rank candidate bundles for each user.

Figure 1 outlines ReCoBR. To address noise in U-B interactions, we first pretrain user and bundle representations, learn a user-conditioned denoiser over historical bundle representations, and use the refined preferences to reconstruct the U-B graph. The denoising objective incorporates a bundle-set consistency loss (SetLoss) to retain bundle-preference information during recovery. To address sparse U-B feedback, we then use the reconstructed U-B view as a common anchor for cross-view contrastive learning before representation fusion. The fused user and bundle representations produce the final recommendation scores.

We distinguish the observed matrix $\mathbf{X}$ from the reconstructed matrix $\tilde{\mathbf{X}}$ throughout this section. The former defines users’ training histories and ranking examples; the latter determines the U-B connections used for recommendation graph propagation. Pretraining, preference construction, and graph reconstruction use only training interactions.

3.2 Diffusion-Driven U-B Graph Denoising and Reconstruction

The purpose of this module is to evaluate an observed interaction in the context of the user’s historical bundle preferences. We construct this context in a pretrained representation space, refine it through conditional denoising, and translate the resulting preference vector into a selection of graph connections.

Pretraining and Historical Preference Construction We pretrain LightGCN (He et al. 2020) on the observed U-B graph using Bayesian Personalized Ranking (BPR) (Rendle et al. 2009). The resulting user and bundle representations, denoted by $\bar{\mathbf{e}}_u, \bar{\mathbf{e}}_b \in \mathbb{R}^d$, remain fixed during graph reconstruction. Here, d is the embedding dimension. These representations provide a common space for constructing historical preferences, conditioning the denoiser, and scoring interactions.

For user u , let $\mathcal{N}_u = \{b \in \mathcal{B} : x_{ub} = 1\}$ be the set of historically interacted bundles. We summarize their representations by mean pooling:

$$\mathbf{p}_{u,0} = \frac{1}{|\mathcal{N}_u|} \sum_{b \in \mathcal{N}_u} \bar{\mathbf{e}}_b. \quad (1)$$

The vector $\mathbf{p}_{u,0}$ represents the user’s observed bundle-preference context. Mean pooling accommodates histories

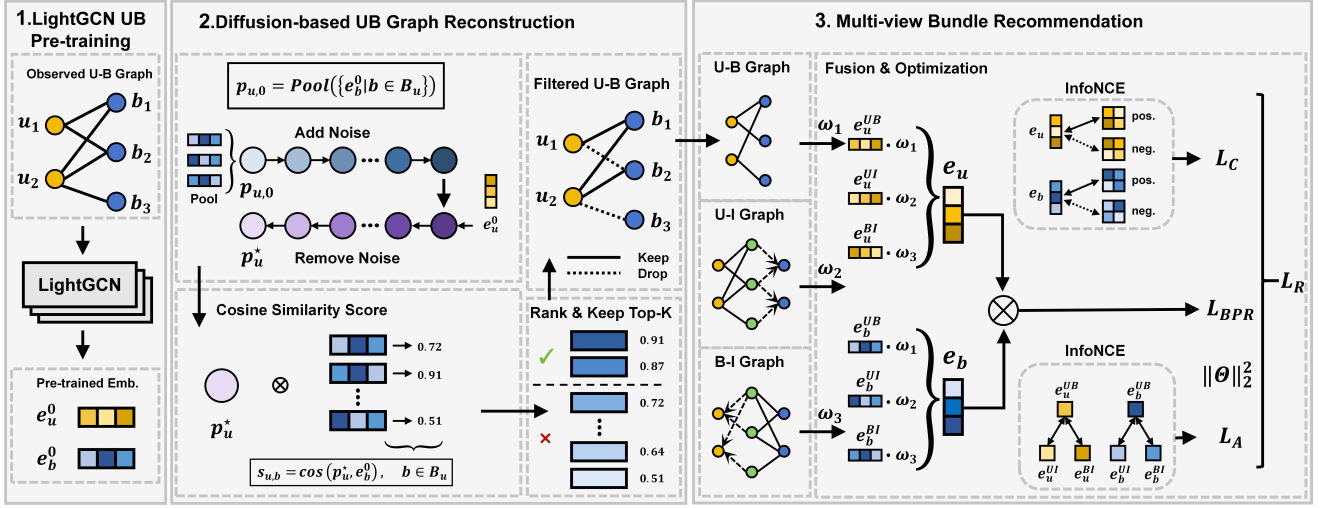


Figure 1: Overall framework of ReCoBR, comprising U-B pretraining, diffusion-driven U-B graph reconstruction, and multi-view recommendation with pre-fusion cross-view contrastive learning.

of different lengths while keeping the denoising input d -dimensional. We construct this vector for users in $\mathcal{U}_+ = \{u \in \mathcal{U} : |\mathcal{N}_u| > 0\}$. The subscript 0 denotes the state before artificial noise is added; it does not assume that the observed history is free of interaction noise.

User-Conditioned Diffusion Denoising We train the denoiser to recover historical preference representations under different levels of perturbation. Following the Gaussian diffusion formulation (Ho, Jain, and Abbeel 2020; Wang et al. 2023), let T be the number of diffusion steps and $\beta_t \in (0, 1)$ the noise variance at step t . We use a linear variance schedule and define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. A perturbed preference vector can be sampled directly as

$$\mathbf{p}_{u,t} = \sqrt{\alpha_t} \mathbf{p}_{u,0} + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d). \quad (2)$$

Here, $\mathbf{I}_d$ is the d -dimensional identity matrix. The cumulative coefficient $\bar{\alpha}_t$ controls how much historical information remains at each noise level.

To make recovery personalized, we condition the denoiser on the pretrained user representation $\bar{\mathbf{e}}_u$. We implement the denoiser f_θ as a multilayer perceptron (MLP), with parameters θ , that predicts the unperturbed preference vector:

$$\begin{aligned} \hat{\mathbf{p}}_{u,0} &= f_\theta(\mathbf{p}_{u,t}, \bar{\mathbf{e}}_u, t), \\ f_\theta(\mathbf{p}, \mathbf{c}, t) &= \text{MLP}_\theta([\mathbf{p}; \mathbf{c}; |\mathbf{p} - \mathbf{c}|; \psi(t)]). \end{aligned} \quad (3)$$

In Eq. (3), $[\cdot; \cdot]$ denotes concatenation, $\odot$ is the element-wise product, and the absolute difference is also element-wise. The vector $\psi(t) \in \mathbb{R}^{d_t}$ is a sinusoidal timestep embedding. The product and difference expose coordinate-wise agreement and discrepancy between the current preference vector and the user condition. Together with the timestep embedding, they allow the MLP to adapt recovery to both the user and the perturbation level. The input dimension is $4d + d_t$, and the output dimension is d .

During training, we uniformly sample $u \in \mathcal{U}_+$ and $t \in \{1, \dots, T\}$ and draw Gaussian noise as in Eq. (2). The reconstruction objective is

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_{u,t,\epsilon} \left[\frac{1}{d} \|\hat{\mathbf{p}}_{u,0} - \mathbf{p}_{u,0}\|_2^2 \right]. \quad (4)$$

This objective learns a shared recovery function from perturbed historical preferences. The original representation supplies the reconstruction target, while the pretrained user embedding provides personalized context.

Bundle-Set Consistency in Denoising Reconstructing the pooled vector provides a basic denoising objective, but subsequent interaction selection depends on the recovered vector’s compatibility with individual historical bundles. We therefore introduce SetLoss to connect preference recovery with the bundle representations that will be used for graph reconstruction. Figure 2 illustrates how SetLoss complements the reconstruction objective in denoiser training.

Let $\text{sim}(\mathbf{a}, \mathbf{b}) = \mathbf{a}^\top \mathbf{b} / (\|\mathbf{a}\|_2 \|\mathbf{b}\|_2)$ denote cosine similarity, with a small numerical safeguard for zero norms. SetLoss encourages average directional agreement between the recovered preference and the user’s historical bundle representations:

$$\mathcal{L}_{\text{set}} = \mathbb{E}_{u,t,\epsilon} \left[\frac{1}{|\mathcal{N}_u|} \sum_{b \in \mathcal{N}_u} (1 - \text{sim}(\hat{\mathbf{p}}_{u,0}, \bar{\mathbf{e}}_b)) \right]. \quad (5)$$

The two objectives emphasize different aspects of recovery. Equation (4) matches the coordinates of the pooled representation, whereas Eq. (5) measures agreement with each historical bundle after vector normalization. This keeps the recovered preference connected to the historical bundle set in the same similarity geometry used for interaction scoring. Averaging over each user’s history also prevents the loss from increasing merely because that user has more observed interactions.

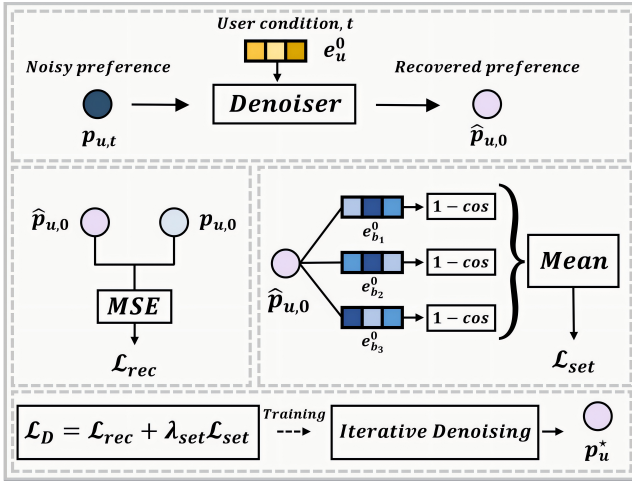


Figure 2: SetLoss in user-conditioned denoising. The recovered preference is compared with the pooled history by MSE and with individual historical bundles by cosine distance, averaged over the history. The joint loss trains the denoiser; subsequent iterative denoising starts from $\mathbf{p}_{u,0}$ to obtain $\mathbf{p}_u^*$. The figure uses $\mathbf{e}_u^0, \mathbf{e}_b^0$ for the fixed pretrained embeddings $\bar{\mathbf{e}}_u, \bar{\mathbf{e}}_b$.

We optimize the denoiser with

$$\mathcal{L}_D = \mathcal{L}_{rec} + \lambda_{set} \mathcal{L}_{set}, \quad (6)$$

where $\lambda_{set} \geq 0$ controls the consistency constraint. SetLoss serves as a task-specific regularizer within the denoising framework: it encourages preservation of bundle-preference context while the reconstruction term learns recovery from perturbations.

Preference Refinement and Graph Reconstruction After learning the denoiser, we use it to refine the historical preference vector for interaction scoring. This stage employs a deterministic refinement procedure initialized with $\mathbf{p}_{u,0}$. Specifically,

$$\begin{aligned} \mathbf{r}_{u,T} &= \mathbf{p}_{u,0}, \\ \mathbf{r}_{u,t-1} &= f_{\theta}(\mathbf{r}_{u,t}, \bar{\mathbf{e}}_u, t), \quad t = T, T-1, \dots, 1, \\ \mathbf{p}_u^* &= \mathbf{r}_{u,0}. \end{aligned} \quad (7)$$

The states $\mathbf{r}_{u,t}$ are refinement iterates, distinct from the forward-corrupted training states $\mathbf{p}_{u,t}$. Each iteration feeds the current estimate back into the denoiser with the next timestep condition. Thus, graph reconstruction applies the learned recovery network to an observed preference context, without sampling a Gaussian starting state.

We measure how closely each observed bundle matches the refined preference:

$$s_{ub} = \text{sim}(\mathbf{p}_u^*, \bar{\mathbf{e}}_b), \quad b \in \mathcal{N}_u. \quad (8)$$

For each user, we retain the highest-scoring $\kappa_u = \min(\kappa, |\mathcal{N}_u|)$ bundles, where the positive integer κ controls the maximum retained degree:

$$\begin{aligned} \tilde{\mathcal{N}}_u &= \text{TopK}(s_{ub}, \kappa_u), \\ \tilde{x}_{ub} &= \mathbf{1}[b \in \tilde{\mathcal{N}}_u]. \end{aligned} \quad (9)$$

Here, TopK returns bundle indices and $\mathbf{1}[\cdot]$ is the indicator function. Users with at most κ observed bundles retain their full histories; users outside $\mathcal{U}_+$ have no retained U-B edges. The binary matrix $\tilde{\mathbf{X}}$ defines the reconstructed U-B graph. Selection is restricted to observed interactions, so this step changes the graph connections used for propagation rather than assigning new interaction labels. Through Eqs. (1)–(9), pretrained collaborative information, personalized denoising, and bundle-set consistency jointly support preference-based graph reconstruction.

3.3 Pre-Fusion Cross-View Contrastive Learning

With the U-B graph reconstructed, we next learn representations from this graph together with user-item behavior and bundle composition. The auxiliary relations provide complementary evidence when a user’s bundle feedback is limited. We first construct user and bundle representations in each view, and then connect them through U-B-anchored contrastive learning before fusion.

View-Specific Representations Following the three-view backbone of MultiCBR (Ma et al. 2024), we construct the U-B, U-I, and B-I propagation graphs from $\tilde{\mathbf{X}}$, $\mathbf{Y}$, and $\mathbf{Z}$, respectively. Each encoder applies L layers of LightGCN-style neighborhood propagation (He et al. 2020) and aggregates the layer outputs. The encoders share trainable initial embeddings for the same entity; these are separate from the fixed pretrained embeddings in Section 3.2.

U-B view. Propagation on the reconstructed U-B graph directly produces user representations $\mathbf{e}_u^{\text{UB}}$ and bundle representations $\mathbf{e}_b^{\text{UB}}$, capturing collaborative information through the retained interactions.

U-I view. Propagation on the U-I graph produces user representations $\mathbf{e}_u^{\text{UI}}$ and item representations $\mathbf{e}_i^{\text{UI}}$. To obtain a bundle representation $\mathbf{e}_b^{\text{UI}}$ in this view, we average the representations of its constituent items according to $\mathbf{Z}$.

B-I view. Propagation on the B-I graph produces bundle representations $\mathbf{e}_b^{\text{BI}}$ and item representations $\mathbf{e}_i^{\text{BI}}$. To obtain a user representation $\mathbf{e}_u^{\text{BI}}$ in this view, we average the representations of the items that the user has interacted with according to $\mathbf{Y}$.

The two item-pooling operations are

$$\mathbf{e}_b^{\text{UI}} = \frac{1}{|\mathcal{I}_b|} \sum_{i \in \mathcal{I}_b} \mathbf{e}_i^{\text{UI}}, \quad \mathbf{e}_u^{\text{BI}} = \frac{1}{|\mathcal{I}_u|} \sum_{i \in \mathcal{I}_u} \mathbf{e}_i^{\text{BI}}, \quad (10)$$

where $\mathcal{I}_b = \{i : z_{bi} = 1\}$ contains the items in bundle b , and $\mathcal{I}_u = \{i : y_{ui} = 1\}$ contains the items interacted with by user u . An empty neighborhood contributes a zero vector.

This step yields three representations for each user, $\{\mathbf{e}_u^{\text{UB}}, \mathbf{e}_u^{\text{UI}}, \mathbf{e}_u^{\text{BI}}\}$, and each bundle, $\{\mathbf{e}_b^{\text{UB}}, \mathbf{e}_b^{\text{UI}}, \mathbf{e}_b^{\text{BI}}\}$, all in $\mathbb{R}^d$. These unfused representations are used to construct the cross-view objective below and are later fused for prediction.

Cross-View Objectives Anchored in U-B Using these view-specific representations, we pair each user’s U-B representation with its U-I and B-I counterparts, and do the same for each bundle. The reconstructed U-B view thus serves as the common reference. Each same-entity pair is a positive pair, while other entities of the same type provide in-batch negatives.

To express the objective compactly, consider two representation matrices $\mathbf{A}$ and $\mathbf{B}$ with m corresponding rows $\mathbf{a}_j$ and $\mathbf{b}_j$. We use the InfoNCE objective (van den Oord, Li, and Vinyals 2018):

$$\ell_\tau(\mathbf{A}, \mathbf{B}) = -\frac{1}{m} \sum_{j=1}^m \log \frac{\exp(\text{sim}(\mathbf{a}_j, \mathbf{b}_j)/\tau)}{\sum_{k=1}^m \exp(\text{sim}(\mathbf{a}_j, \mathbf{b}_k)/\tau)}. \quad (11)$$

The temperature $\tau > 0$ controls sensitivity to similarity differences. The corresponding row is the positive candidate; all other rows are negative candidates. The denominator includes the positive candidate.

Let $\mathcal{S}_U$ and $\mathcal{S}_B$ be the sets of unique users and unique positive bundles in a recommendation minibatch. For view $v \in \{\text{UB}, \text{UI}, \text{BI}\}$, let $\mathbf{E}_U^v$ and $\mathbf{E}_B^v$ contain the representations of these users and bundles, respectively, with a consistent entity order across views. The pre-fusion objective is

$$\begin{aligned} \mathcal{L}_A = & \ell_{\tau_a}(\mathbf{E}_U^{\text{UB}}, \mathbf{E}_U^{\text{UI}}) + \ell_{\tau_a}(\mathbf{E}_U^{\text{UB}}, \mathbf{E}_U^{\text{BI}}) \\ & + \ell_{\tau_a}(\mathbf{E}_B^{\text{UB}}, \mathbf{E}_B^{\text{UI}}) + \ell_{\tau_a}(\mathbf{E}_B^{\text{UB}}, \mathbf{E}_B^{\text{BI}}). \end{aligned} \quad (12)$$

Here, τ_a is the cross-view temperature. Deduplication prevents repeated occurrences of the same entity from being treated as negatives in this objective. A term is skipped if its entity set contains fewer than two members.

Equation (12) encourages agreement for the same user or bundle across U-B–U-I and U-B–B-I pairs while preserving discrimination between entities. The U-B view is the common anchor in constructing the objective, and gradients update both representations in each pair. These constraints allow item-level behavioral and compositional evidence to participate directly in bundle-preference learning, providing additional training signals beyond an individual user’s limited U-B history.

3.4 Prediction and Model Optimization

View fusion and prediction. We combine the three view-specific representations into a single user representation and a single bundle representation. Their inner product gives the preference score:

$$\begin{aligned} \mathbf{e}_u &= \omega_1 \mathbf{e}_u^{\text{UB}} + \omega_2 \mathbf{e}_u^{\text{UI}} + \omega_3 \mathbf{e}_u^{\text{BI}}, \\ \mathbf{e}_b &= \omega_1 \mathbf{e}_b^{\text{UB}} + \omega_2 \mathbf{e}_b^{\text{UI}} + \omega_3 \mathbf{e}_b^{\text{BI}}, \\ \hat{y}_{ub} &= \mathbf{e}_u^\top \mathbf{e}_b. \end{aligned} \quad (13)$$

Here, ω_1, ω_2 , and ω_3 control the contributions of the U-B, U-I, and B-I views, respectively. They are nonnegative and satisfy $\omega_1 + \omega_2 + \omega_3 = 1$. The same coefficients are used for

users and bundles. They are dataset-specific hyperparameters held fixed during training, rather than learnable parameters.

Recommendation objectives. Representation fusion and the fused-space contrastive regularizer are components of the recommendation backbone (Ma et al. 2024). Let $\mathbf{H}_U$ and $\mathbf{H}_B$ contain the fused user and positive-bundle representations in minibatch order. The backbone self-contrastive regularizer used here is

$$\mathcal{L}_C = \frac{1}{2} [\ell_{\tau_c}(\mathbf{H}_U, \mathbf{H}_U) + \ell_{\tau_c}(\mathbf{H}_B, \mathbf{H}_B)], \quad (14)$$

where τ_c is its temperature. Each row is paired with itself in the numerator, and the denominator contains all rows of the corresponding minibatch matrix. This term regularizes separation in the fused representation space; $\mathcal{L}_A$ explicitly connects the different relation views before fusion.

For a minibatch $\mathcal{Q}$ of ranking triplets (u, b^+, b^-) , the positive bundle satisfies $x_{ub^+} = 1$ and the negative bundle is sampled from $\mathcal{B} \setminus \mathcal{N}_u$. We optimize BPR (Rendle et al. 2009) together with the two regularization objectives:

$$\begin{aligned} \mathcal{L}_{\text{BPR}} &= -\frac{1}{|\mathcal{Q}|} \sum_{(u, b^+, b^-) \in \mathcal{Q}} \log \sigma(\hat{y}_{ub^+} - \hat{y}_{ub^-}), \\ \mathcal{L}_R &= \mathcal{L}_{\text{BPR}} + \lambda_a \mathcal{L}_A + \lambda_c \mathcal{L}_C + \lambda_r \|\Theta\|_2^2. \end{aligned} \quad (15)$$

Here, σ is the sigmoid function, Θ comprises the trainable initial user, bundle, and item embeddings of the recommendation backbone, and $\lambda_a, \lambda_c, \lambda_r \geq 0$ balance the loss terms. Ranking examples continue to come from $\mathbf{X}$, even when their edges are not selected for $\tilde{\mathbf{X}}$. Graph reconstruction therefore filters propagation connections while preserving the observed feedback used for ranking.

Training and inference. In implementing the graph encoders, we use symmetrically normalized propagation and fixed, view-specific layer coefficients to combine the initial embeddings with row-normalized propagated outputs. Following the backbone’s noise augmentation setting (Ma et al. 2024), we apply small representation perturbations during propagation and auxiliary item pooling in training, and disable them at evaluation.

Training alternates between graph reconstruction and recommendation learning. After offline pretraining, each epoch first updates the denoiser using $\mathcal{L}_D$ over the active users. We then reconstruct $\tilde{\mathbf{X}}$ with the updated denoiser and optimize the recommendation backbone using $\mathcal{L}_R$ over ranking minibatches. The denoiser and backbone have separate optimizers; the reconstructed graph is held fixed during the recommendation updates within an epoch. Discrete edge selection does not propagate ranking gradients back into the denoiser.

For recommendation, we compute and cache the final fused embeddings using the reconstructed graph associated with the selected model. Candidate bundles are ranked by Eq. (13), with training-observed bundles excluded. The denoiser is used for graph reconstruction and is not evaluated separately for each candidate bundle.

Dataset	#Users	#Items	#Bundles	#U-I	#U-B	#Avg. I/B
Youshu	8,039	32,770	4,771	138,515	51,377	37.03
NetEase	18,528	123,628	22,864	1,128,065	302,303	77.80
iFashion	53,897	42,563	27,694	2,290,645	1,679,708	3.86

Table 1: Dataset statistics. U-I and U-B count user-item and user-bundle interactions; Avg. I/B is the average number of items per bundle.

3.5 Computational Complexity

Let $M = |\mathcal{E}_{UB}|$ be the number of observed U-B edges, and let $E = |\tilde{\mathcal{E}}_{UB}| + |\mathcal{E}_{UI}| + |\mathcal{E}_{BI}|$ be the total number of edges used by the three propagation graphs. Let Q be the number of recommendation minibatches per epoch, m their maximum size, and C_f the cost of a denoiser evaluation for one user. For a fixed-depth MLP whose hidden and timestep dimensions are proportional to d , $C_f = O(d^2)$.

With one denoiser-training pass and one graph refresh per epoch, the main time cost is

$$O\left((T+1)|\mathcal{U}_+|C_f + M(d + \log(1 + \kappa)) + Q(LEd + m^2d)\right). \quad (16)$$

The first term accounts for denoiser training and iterative preference refinement. The second includes history pooling, SetLoss evaluation, observed-edge scoring, and bounded top- κ selection. The remaining terms cover graph propagation and minibatch contrastive computations. Offline LightGCN pretraining adds a graph-propagation cost of $O(LMd)$ per propagation pass. Since $|\tilde{\mathcal{E}}_{UB}| \leq M$, graph reconstruction does not densify the propagation graph or require scoring all $|\mathcal{U}||\mathcal{B}|$ pairs.

The trainable parameters occupy $O((|\mathcal{U}| + |\mathcal{B}| + |\mathcal{I}|)d + P_\theta)$ space, where P_θ is the denoiser parameter count. Fixed pretrained user and bundle embeddings additionally require $O((|\mathcal{U}| + |\mathcal{B}|)d)$ storage. Training also stores sparse graphs, intermediate graph representations, and temporary minibatch buffers. Once fused embeddings are cached, scoring one user-bundle pair costs $O(d)$.

4 Experiments

We evaluate ReCoBR on three bundle recommendation benchmarks. The experiments examine its overall ranking performance, the contributions of graph reconstruction and pre-fusion cross-view learning, and their behavior under different interaction selection strategies and levels of user feedback. We also analyze sensitivity to the key hyperparameters.

4.1 Experimental Settings

Datasets and Evaluation. We use Youshu, NetEase, and iFashion for book-list, music-playlist, and outfit recommendation, respectively. Each dataset contains U-B interactions, U-I interactions, and B-I affiliations. Table 1 summarizes their statistics. We adopt the preprocessed datasets and splits used by CrossCBR and MultiCBR (Ma et al. 2022, 2024),

with 70%, 10%, and 20% of U-B interactions assigned to training, validation, and testing. Pretraining, historical preference construction, and U-B graph reconstruction use only the training interactions.

We report Recall@ K and NDCG@ K for $K \in \{10, 20\}$ under full-ranking evaluation. For each user, we rank all candidate bundles after excluding those observed in training. The checkpoint with the best Top-20 validation performance is selected for test evaluation.

Baselines. We compare ReCoBR with seven bundle recommendation methods. BGCN (Chang et al. 2020) models bundle-level and item-level collaborative information through graph propagation, while MIDGN (Zhao et al. 2022) disentangles user intents across global and local views. CrossCBR (Ma et al. 2022) connects bundle-view and item-view representations through contrastive learning, and BundleGT (Wei et al. 2023) uses a hierarchical graph Transformer to model bundling strategies. MultiCBR (Ma et al. 2024) encodes all three relations and applies contrastive learning after view fusion. EBRec (Du et al. 2025) enhances item-level bundle representations through bundle-user-item correlations. EpicCBR (Li, Liu, and Wei 2026) combines item-relation enhancement with contrastive learning across warm-start and cold-start scenarios.

For MultiCBR, EBRec, and EpicCBR, we run their released implementations on the same splits and evaluation protocol as ReCoBR and report averages over three independent runs. ReCoBR is evaluated in the same manner. The BGCN, MIDGN, and CrossCBR results are taken from the comparisons reported in MultiCBR (Ma et al. 2024); the BundleGT results are taken from its published evaluation. Missing results are left unreported in Table 2.

Implementation Details. We implement ReCoBR in PyTorch and train it on a single NVIDIA GeForce RTX 3090 GPU. The recommendation embeddings use Xavier initialization, dimension 64, and two graph propagation layers. We use Adam with a learning rate of 10^{-3} , a recommendation batch size of 2048, and one negative bundle per positive interaction. Ranking examples are drawn from the original training interactions; the reconstructed U-B graph determines the propagation connections.

The denoiser is a two-layer MLP with hidden dimension 128, timestep embedding dimension 16, and dropout rate 0.1. Its separate optimizer uses a learning rate of 10^{-3} , weight decay 10^{-5} , and batch size 1024. We use a linear noise schedule from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. In the order Youshu, NetEase, and iFashion, the diffusion steps T are (9, 12, 12), the retention limits κ are (7, 13, 3), and the SetLoss weights λ_{set} are (0.04, 0.05, 0.05).

The fixed fusion coefficients $(\omega_1, \omega_2, \omega_3)$, corresponding to U-B, U-I, and B-I, are (0.35, 0.15, 0.50) for Youshu, (0.50, 0.20, 0.30) for NetEase, and (0.00, 0.50, 0.50) for iFashion. A zero U-B fusion coefficient removes its direct contribution to the final weighted sum; the U-B view still participates in pre-fusion contrastive learning. In the same dataset order, the pre-fusion loss weights λ_a are (0.004, 0.005, 0.07), and the post-fusion loss weights λ_c are (0.04, 0.05, 0.10). Both contrastive temperatures are 0.23 on

Method	Youshu				NetEase				iFashion			
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
BGCN	–	–	0.2436	0.1329	–	–	0.0626	0.0328	–	–	0.0802	0.0582
MIDGN	–	–	0.2682	0.1527	–	–	0.0678	0.0343	–	–	0.0856	0.0473
CrossCBR	–	–	0.2823	0.1670	–	–	0.0844	0.0458	–	–	0.1132	0.0872
BundleGT	–	–	<u>0.2905</u>	<u>0.1737</u>	–	–	0.0903	0.0478	–	–	0.1263	0.0969
MultiCBR	0.1995	0.1439	0.2847	0.1681	0.0578	0.0389	0.0901	0.0491	<u>0.1057</u>	<u>0.1028</u>	<u>0.1494</u>	<u>0.1203</u>
EBRec	0.1984	0.1425	0.2797	0.1659	0.0559	0.0376	0.0881	0.0477	0.0909	0.0874	0.1327	0.1044
EpicCBR	<u>0.1999</u>	<u>0.1443</u>	0.2835	0.1682	<u>0.0601</u>	<u>0.0397</u>	<u>0.0926</u>	<u>0.0503</u>	0.1049	0.1022	0.1467	0.1190
ReCoBR	0.2053	0.1482	0.2920	0.1759	0.0617	0.0411	0.0955	0.0511	0.1144	0.1146	0.1568	0.1315
Improv.	2.70%	2.70%	0.52%	1.27%	2.66%	3.53%	3.13%	1.59%	8.23%	11.48%	4.95%	9.31%

Table 2: Overall performance on three datasets. R and N denote Recall and NDCG. Bold and underlined entries mark the best and second-best reported results, respectively. Improv. is ReCoBR’s relative improvement over the strongest baseline for each metric; dashes indicate unreported results.

Variant	Youshu		NetEase		iFashion	
	R@20	N@20	R@20	N@20	R@20	N@20
ReCoBR	0.2920	0.1759	0.0955	0.0511	0.1568	0.1315
w/o Graph Recon.	0.2919	0.1734	0.0948	0.0509	0.1500	0.1245
w/o SetLoss	0.2913	0.1749	0.0951	0.0506	0.1559	0.1301
w/o All CL	0.2326	0.1307	0.0512	0.0260	0.0400	0.0281
w/o Pre-fusion CL	0.2901	0.1692	0.0906	0.0487	0.1558	0.1304
w/o Post-fusion CL	0.2595	0.1502	0.0682	0.0350	0.0461	0.0350

Table 3: Component ablations on three datasets, evaluated by Recall@20 (R@20) and NDCG@20 (N@20). Bold entries identify the complete ReCoBR model.

Youshu and 0.20 on NetEase and iFashion. The embedding regularization coefficient λ_r is 10^{-5} . Hyperparameters are selected using validation performance.

4.2 Overall Performance

Table 2 compares ReCoBR with the seven baselines. ReCoBR obtains the highest reported Recall and NDCG at both cutoffs on all three datasets. The strongest baseline varies: EpicCBR leads at Top-10 on Youshu and on all four metrics on NetEase, BundleGT leads at Top-20 on Youshu, and MultiCBR leads on iFashion. ReCoBR improves over each of these methods in its respective strongest setting.

The gains are largest on iFashion, with relative improvements of 8.23%, 11.48%, 4.95%, and 9.31% in R@10, N@10, R@20, and N@20, respectively. At both cutoffs, the larger gains in NDCG than in Recall indicate that the improvement includes placing relevant bundles higher in the

recommendation list. On Youshu and NetEase, the margins are smaller but remain positive across all four metrics.

The comparison with MultiCBR is particularly informative because it already uses the three relation views, representation fusion, and post-fusion contrastive learning. ReCoBR improves over this baseline across all datasets and metrics. This motivates examining the additional graph reconstruction and pre-fusion objectives separately, as we do next, to understand their contributions beyond the multi-view backbone.

4.3 Ablation Study

Table 3 compares ReCoBR with five variants. **w/o Graph Recon.** replaces the reconstructed U-B graph with the original observed graph while retaining the recommendation objectives. **w/o Pre-fusion CL** removes the U-B-anchored cross-view objective but retains graph reconstruction. **w/o Post-fusion CL** removes the contrastive regularizer in the fused representation space, and **w/o All CL** removes both contrastive objectives.

Diffusion-Driven Graph Reconstruction. Removing graph reconstruction reduces R@20 and N@20 on iFashion from 0.1568 and 0.1315 to 0.1500 and 0.1245, respectively. Both metrics also decrease on NetEase. On Youshu, R@20 is essentially unchanged (0.2920 versus 0.2919), whereas N@20 drops from 0.1759 to 0.1734. Thus, the benefit of reconstruction varies across datasets and metrics, with the clearest gains on iFashion. These results support refining the U-B connections used for propagation through the complete pretraining, conditional denoising, and interaction-selection pipeline.

Bundle-Set Consistency. We additionally examine **w/o SetLoss**, which removes the bundle-set consistency term from denoiser training while retaining the reconstruction objective and both ReCoBR modules. Removing this term yields a modest performance decrease across all three datasets. This supports its role as an auxiliary task-specific constraint: maintaining agreement with historical bundle representations helps the recovered preference serve subsequent interaction scoring.

Pre-Fusion Cross-View Learning. Removing the pre-fusion objective lowers both metrics on all three datasets. On NetEase, R@20 and N@20 decrease from 0.0955 and 0.0511 to 0.0906 and 0.0487, respectively; on Youshu, N@20 falls from 0.1759 to 0.1692. These comparisons show that explicit U-B–U-I and U-B–B-I associations provide value beyond representation fusion and its existing regularizer. Using the reconstructed U-B view as a common reference allows auxiliary behavioral and compositional information to contribute to bundle-preference learning before the views are combined.

Backbone Contrastive Regularization. Removing post-fusion contrastive learning causes larger decreases, and removing both contrastive objectives further reduces performance on every dataset. The fused-space regularizer therefore remains important to the inherited recommendation backbone. Meanwhile, the full model’s gains over **w/o Pre-fusion CL** show that this regularizer does not make the proposed cross-view objective redundant. The two objectives constrain different representations: relation-specific pairs before fusion and the combined representations used for prediction.

4.4 In-depth Analysis

Interaction Selection Strategies. To examine how the choice of retained interactions affects recommendation, we compare five U-B graph construction strategies on iFashion. **Original UB** keeps the observed graph. **Random- K** retains randomly selected observed interactions, whereas **Popularity- K** selects those associated with the most popular bundles. **Pretrain-Sim** scores interactions by the cosine similarity of pretrained user and bundle embeddings. **ReCoBR** instead scores them against the user’s refined historical bundle-preference vector. Each filtering strategy retains at most κ interactions per user, where κ is denoted by K in the figure, and selects only from the user’s training history. Results are averaged over multiple independent runs.

Figure 3(a)–(b) shows that ReCoBR achieves the highest Recall@20 and NDCG@20 among the compared strategies at every tested retention limit. Random selection can improve over the original graph, but consistently trails ReCoBR at matched retention limits. Thus, the choice of connections matters beyond reducing their number. Popularity-based selection is especially weak in NDCG, indicating that globally popular bundles are not necessarily the most useful propagation neighbors for an individual user.

Pretrain-Sim also underperforms ReCoBR. This comparison favors the complete history-based refinement and scoring pipeline over direct pretrained user–bundle similarity.

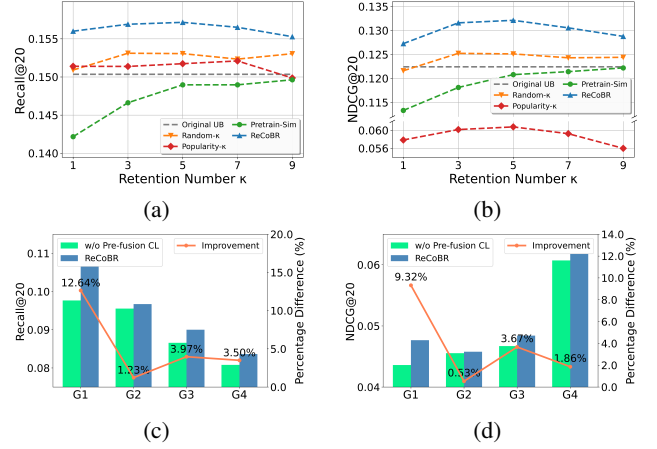


Figure 3: **In-depth analysis of ReCoBR.** (a)–(b) Recall@20 and NDCG@20 of interaction selection strategies on iFashion, with retention limit κ plotted as K . (c)–(d) Recall@20 and NDCG@20 of complete ReCoBR and w/o Pre-fusion CL on NetEase. Both retain graph reconstruction. G1–G4 are ordered by training U-B degree, with ranges $[0, 7)$, $[7, 8)$, $[8, 12)$, and $[12, 660]$. Lines show ReCoBR’s relative gains over w/o Pre-fusion CL.

ReCoBR’s performance initially improves as more interactions are retained and then declines at larger limits. Together, these observations support selecting connections according to personalized bundle-preference context while balancing neighborhood coverage and selectivity.

Performance on Sparse Users. To evaluate the pre-fusion objective under limited bundle feedback, we group NetEase users by their numbers of training U-B interactions, from the lowest-degree group G1 to the highest-degree group G4. The degree ranges are given in Figure 3. We compare complete ReCoBR with **w/o Pre-fusion CL**. Both settings retain diffusion-driven graph reconstruction and the recommendation backbone; they differ in whether the U-B-anchored cross-view objective is included.

As shown in Figure 3(c)–(d), ReCoBR improves Recall@20 and NDCG@20 in all four groups. The largest relative gains occur in G1: 12.64% in Recall@20 and 9.32% in NDCG@20. Gains in the other groups are smaller and do not decrease monotonically with interaction count. This result directly supports the role of pre-fusion cross-view learning under sparse U-B feedback. By associating U-B representations with representations derived from U-I behavior and B-I composition, the objective provides additional learning signals for users with limited bundle histories. The gains for G2–G4 further indicate that its benefit is not restricted to the lowest-interaction users.

4.5 Hyperparameter Sensitivity

We vary the maximum diffusion noise variance β_T , the retention limit κ , and the pre-fusion contrastive weight λ_a one at a time while keeping the other settings fixed. Figure 4 reports Recall@20.

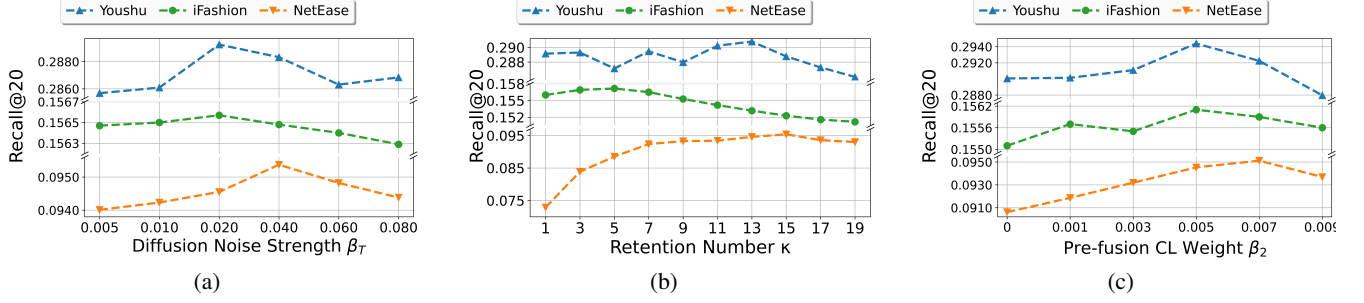


Figure 4: Hyperparameter sensitivity measured by Recall@20 on three datasets. (a) Maximum per-step noise variance β_T in the linear diffusion schedule. (b) Retention limit κ for U-B graph reconstruction. (c) Pre-fusion contrastive weight λ_a , labeled β_2 in the plot. Broken y -axes separate the performance scales of the three datasets.

Diffusion Noise. Figure 4(a) shows that intermediate noise levels perform better than the ends of the tested range. Youshu and NetEase exhibit clearer changes, whereas iFashion varies within a narrower absolute range. Increasing β_T makes the recovery task harder by perturbing the historical preference more strongly. The observed curves are consistent with balancing this training perturbation against preservation of personalized information; stronger corruption does not consistently improve recommendation.

Retention Limit. Figure 4(b) reveals different responses to κ . NetEase benefits markedly from increasing a very small retained neighborhood, then levels off and slightly declines. iFashion favors smaller limits and loses accuracy as more connections are included. Youshu fluctuates within a narrower range. These differences show that aggressive filtering can discard useful propagation connections, while retaining more observed edges is not always beneficial. The retention limit should therefore account for the dataset’s balance between neighborhood coverage and preference relevance.

Pre-Fusion Contrastive Weight. In Figure 4(c), moderate nonzero weights improve over omitting the cross-view objective, especially on NetEase. Larger weights eventually reduce performance; on Youshu, the largest tested weight also underperforms zero weight. This behavior highlights the need to balance cross-view agreement with the ranking objective, rather than maximizing alignment strength. The scale of these changes differs across datasets, as indicated by the broken axes.

5 Conclusion

We presented ReCoBR for bundle recommendation with noisy and sparse U-B feedback. Its central design connects pretraining, user-conditioned latent diffusion, and interaction selection to reconstruct the graph used for recommendation. Within this pipeline, bundle-set consistency adapts preference recovery to the historical bundles used for subsequent scoring. Pre-fusion cross-view contrastive learning then connects representations from the reconstructed U-B graph with the U-I and B-I views on both the user and bundle sides.

Experiments on three public benchmarks show improvements over the compared baselines. Component ablations

and interaction selection analyses support the graph reconstruction pipeline and the additional cross-view objective. The user-group analysis further shows that pre-fusion learning yields its largest relative gains for users with the fewest bundle interactions. Together, these findings support refining the U-B propagation graph and using its representations to guide auxiliary-view learning as a way to improve bundle recommendation under noisy and limited feedback.

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